RepliVision Inc.

School of Engineering Science
Simon Fraser University
British Columbia, V5A 1S6

Contact: Mehrshad Azizinia (CEO)
mehrshad_azizinia@sfu.ca

Replivision

Date Printed: October 12, 2025

Team:

Mehrshad Azizinia (CEO)

Curtis Huang (CCO)

Joseph Wen (CPO)

Shervin Khamooshian (CTO)

This document includes:

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Bringing certainty to dining experiences by letting customers preview food exactly how it would be prepared and then order it from any device.

Signatures

Project manager: Azizinia Date: 12/10/2025

Sponsor: ____________ Date: 12/10/2025

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Date

October 12, 2025

Project Title

RepliVision

Authorization

This charter, established between Wayne and Mehrshad Azizinia, authorizes the RepliVision team to commence development of a functional shelf-ready service. The project consists of a 3D food scanning system integrated with a cloud-hosted web application for digital menus and ordering.

Project Objective/Overview

RepliVision addresses the gap in restaurant food presentation by allowing customers to preview menu items before ordering. The system combines:

- *A 3D food scanner to capture accurate, high-fidelity models of plated meals.*
- *A web application hosting scanned dishes in an interactive digital menu.*
- *Easy customer access via restaurant-provided iPads or QR code links.*

This service aims to increase customer trust, reduce order hesitation, and provide restaurants with a modern digital tool that enhances dining experiences while remaining practical and scalable.

High-level requirements

STATEMENT OF WORK

- *Build a functional 3D food scanner.*
- *Develop a cloud-based infrastructure to store and host 3D models.*
- *Design an interactive web application with a menu and ordering features.*
- *Enable cross-platform access via QR codes and tablets.*
- *Deliver ENSC 440 documentation, financials, and final project video*

High-level project description, boundaries, and key deliverables

SCOPE OF WORK

- *Assemble 3D scanning hardware and integrate with reconstruction software.*
- *Implement a model upload and storage pipeline to the web server.*
- *Build a responsive web interface for browsing and ordering food items.*
- *Integrate QR code functionality for customer access.*
- *Conduct testing and usability validation with restaurant-style scenarios.*
- *Provide comprehensive project documentation and deliverables.*

Executive Summary

RepliVision Inc. is redefining the dining experience by combining 3D scanning technology with a digital menu platform. Our service allows customers to preview realistic 3D models of menu items through an interactive digital menu system, accessible on restaurant-provided tablets or personal devices via QR codes.

The Problem

In restaurants, customers often struggle to accurately visualize menu items. Static images or text descriptions fail to capture portion size, presentation, or ingredients, leading to ordering hesitation, dissatisfaction, or food waste. For many small and mid-sized restaurants, professional food photography also poses challenges, it *requires specialized equipment, lighting, and styling expertise, which can be time-consuming and costly to maintain for changing menus. As a result, restaurants frequently rely on low-quality images that fail to engage customers, leading to decreased confidence, longer ordering times, and fewer opportunities to promote premium offerings.*

The Solution

RepliVision offers an integrated solution that simplifies the creation of high-quality menu visuals. Restaurant staff can easily generate realistic images by placing the system onto a plated dish, making the process far quicker and more convenient than traditional food photography.

A 3D food scanner captures a high-fidelity digital model of the meal, which is automatically uploaded to an interactive digital menu. Customers can preview dishes in lifelike 3D on tablets provided by the restaurant or on their own devices using QR codes.

This system removes the need for professional photography, enhances transparency, boosts customer confidence, and creates a modern dining experience aligned with evolving consumer expectations.

The Opportunity

The restaurant industry is highly competitive, with increasing demand for digital transformation and immersive customer experiences. Globally, the digital restaurant ordering market is projected to grow significantly as customers seek convenience and clarity before purchase. By reducing uncertainty and enhancing engagement, RepliVision positions itself as an innovative tool for mid- to high-tier restaurants, fast-casual dining, and hospitality venues.

Competitive Advantage

Unlike traditional photo-based menus or expensive hardware-based holographic solutions, RepliVision offers:

Accuracy: True-to-scale 3D models rather than stylized images.

Accessibility: Cloud-based menus are accessible on existing tablets and smartphones, reducing hardware costs.

Scalability: Easily expandable across multiple restaurants and menus.

Customer Experience: Faster decision-making, higher satisfaction, and potential for increased order value.

The Team

RepliVision Inc. is led by a multidisciplinary team of engineering science students:

Mehrshad Azizinia (CEO, Project Manager*): Project leadership, complete system integration, complete project overview, stakeholder communication, cloud deployment, software integration, system design.*

Shervin Khamooshian (CTO)*: 3D reconstruction algorithms, software integration, cloud deployment.*

Curtis Huang (CCO)*: Documentation, user interface design, and demonstration coordination.*

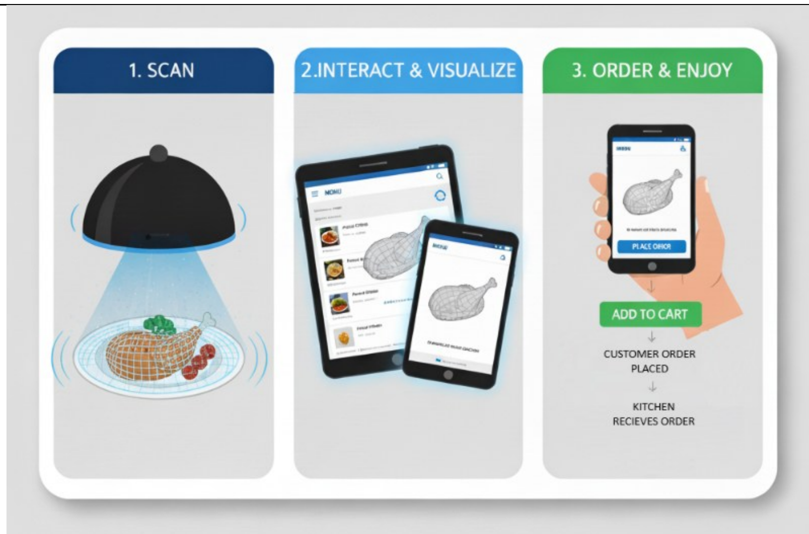
Joseph Wen (CPO)*: Hardware design, sensor calibration, and quality assurance.*

Together, the team combines technical expertise in hardware, software, and user experience design with strong collaborative and project management skills.

Our Ask

RepliVision is seeking support from the ENSC 440 instructional team in the form of guidance, lab access, and resources to refine our prototype into a functional, shelf-ready service. Additionally, modest financial support for hardware procurement (~$540) and cloud infrastructure (~$200) will ensure reliable testing and *deployment. This investment will allow us to deliver a validated system, comprehensive documentation, and a demonstration-ready product by the end of the Capstone course.*

Graphical Abstract



Scope

Product Scope Description
The RepliVision system delivers a compact 3D food scanner paired with an interactive menu platform. The scanner allows restaurant staff to place a plated dish inside, initiate a capture, and automatically generate a true-to-scale 3D image.

Scope Details		
	In Scope	*Must (M), Should (S), Could (C), Won't (W)*
1	*Basic usability testing and documentation (how-to guide for everyday users).*	*(M)*
2	*Design and prototype of a dome-style scanner hardware rig (camera mounts, consistent lighting, structural housing).*	*(M)*
3	*Camera calibration and synchronized image capture workflow.*	*(M)*
4	*Simple operator interface to initiate scanning and monitoring capture progress*	*(M)*
5	*Cloud-hosted web catalogue, a web application for storing, previewing, and managing 3D food models, accessible via tablets.*	*(M)*
6	*Multiple food/dish size accommodation, at least 2 sizes (sanitation considerations, cleanable dome surface).*	*(S)*
7	*Optional lighting optimization experiments for reflective/glossy surfaces.*	*(C)*
8	*QR code generation system enabling customer access from personal smartphones*	*(C)*
9	*Full industrial/commercial certification or mass manufacturing design*	*(W)*

	Out of Scope	*Must (M), Should (S), Could (C), Won't (W)*
1	*HoloVision hologram display or Pepper's Ghost display development.*	*(M)*
2	*Integration with restaurant ordering, POS, or delivery platforms.*	*(S)*
3	*Mobile AR viewer apps or complex AR integration for live table displays.*	*(C)*
4	*Large-scale production, retail packaging, or supply-chain readiness.*	*(W)*
5	*Enterprise-class cloud infrastructure (we will use a minimal-hosted solution for the 440).*	*(W)*

KPIs

KPI Details				
Feature Requirement	*KPI Metric*	*Target Value*	*Measurement Method*	*Relevance*
Basic usability testing and documentation	*Documentation guide clarity*	*>90% success using only documentation*	*Users attempt full scan cycle with only documentation as reference*	*Confirms documentation sufficiency for independent operation*
Scanner (dome) hardware rig	*Structural stability*	*Zero (surface) movement during scan cycle*	*Verification that all cameras produce valid images each scan cycle*	*Ensures consistent imaging geometry for accurate reconstruction*
Scanner (dome) hardware rig	*Lighting consistency*	*≤ 5% luminance variation across capture area*	*Verification that all cameras produce valid images each scan cycle*	*Provides uniform illumination for consistent video quality*
Camera image capture	*Image capture completeness*	*100% successful image acquisition per scan*	*Verification that all cameras produce valid images each scan cycle*	*Confirms no missing data in reconstruction pipeline*
Simple operator interface	*Interface response time*	*≤ 1 second for all user actions*	*Measurement of time between button press and system response/feedback*	*Ensures responsive user experience during operation*
Simple operator interface	*Status feedback clarity*	*100% correct user interpretation of status*	*Users identify current system state (ready/scanning) from interface*	*Validates clear communication of system status*
Cloud-hosted web catalogue	*Model upload success rate*	*≥ 98% successful uploads*	*Ratio of successful model uploads to cloud storage over 10 scan attempts*	*Demonstrates reliable cloud connectivity and data transfer*
Cloud-hosted web catalogue	*Model load time on tablet*	*≤ 3 seconds*	*Browser performance measurement on tablet hardware with 50 Mbps connection*	*Ensures responsive customer browsing experience*
Cloud-hosted web catalogue	*Model preview accuracy*	*100% visual fidelity match*	*Side-by-side comparison confirms web-displayed model matches uploaded source file*	*Validates no corruption or loss during storage/retrieval*

Multiple dish size accommodation	Size compatibility range	Support for 2 standard plate sizes (8" and 10" diameter)	Physical testing with standard restaurant plates; verification of complete capture	Accommodates most common restaurant serving sizes

Project Budget

	Item	Budget
1	3D Scanner Components (Raspberry Pi, Cameras, Motors, Enclosure)	$540
2	Cloud Hosting & Web Services (Servers, Storage, Domain)	$370
3	3D Printing Materials and Enclosure Fabrication	$60
4	Miscellaneous & Testing Materials	$90
5	Contingency Reserve	$100
	Total	$1160

Summary Schedule

	Milestone	Completion Date
1	Hardware Procurement and Assembly	3 weeks
2	3D Scanning Software Development	2 weeks
3	Web Application Development	3 weeks
4	System Integration & Usability Testing	3 weeks
5	Documentation & Final Video Preparation	1 week
	Total	12 weeks

Risk Assessment

	Risk Event	Probability	Impact	Response	Contingency Allowance ($)
1	Scan Accuracy Issues	M	H	Mitigate with calibration routines and backup software.	$100
2	Web Performance Challenges	M	M	Optimize 3D model size; enable cloud scaling.	$40
3	Component Delays	L	H	Procure early; identify alternate suppliers.	$75
4	User Adoption Resistance	M	M	Provide intuitive UI and training materials.	$50
5	Failure to meet Food Safety Regulations	L	H	Conduct regulatory compliance review; integrate food-grade materials and proper sanitation protocols	$125
				Total	$400

Roles & Responsibilities

Project Sponsor (ENSC 440 Instructional Team)

1. *Is accountable for the overall achievement of the approved ENSC 440 project objectives*
2. *Provides project requirements and academic standards for capstone deliverables*
3. *Establishes project scope, timeline, and assessment criteria, including technical specifications*
4. *Reviews and approves the project charter, authorizing the commencement of development work*
5. *Monitors progress through scheduled reviews and milestone assessments*
6. *Provides guidance on technical challenges and resource allocation decisions*
7. *Evaluates final deliverables, including functional prototype and comprehensive documentation*

Project Manager (Mehrshad Azizinia - CEO)

1. *Provides overall project leadership and coordinates team activities within the established scope*

2. *Determines a detailed implementation plan, including task assignments and resource allocation*
3. *Establishes organizational structure and assigns specific responsibilities to team members*
4. *Manages procurement of components and materials within approved budget constraints*
5. *Coordinates the software integration of the project*
6. *Resolves technical conflicts and manages stakeholder communications*
7. *Authorizes scope changes within project boundaries and secures approval for major modifications*

Team Member Responsibilities:

1. ***Curtis Huang (CCO):*** *Project documentation, user interface design, demonstration coordination*
2. ***Shervin Khamooshian (CTO):*** *3D reconstruction algorithms, software integration*
3. ***Joseph Wen (CPO):*** *Hardware design, embedded systems, sensor calibration, quality assurance*

Lessons Learned Register

Category	*Issue*	*Problem/Success*	*Impact*	*Recommendation*
Technical	*3D Reconstruction Algorithm Selection*	*Underestimated computational complexity for real-time processing requirements*	*High*	*Allocate additional development time for software optimization and consider hybrid processing approaches to balance presentation and performance*
Team	*Inter-disciplinary Communication*	*Successfully established regular technical synchronization meetings*	*Med*	*Continue structured weekly reviews with shared documentation standards for ENSC 440*
Instructional	*Scope Management Balance*	*Achieved an effective balance between an ambitious technical vision and a realistic deliverable timeline*	*High*	*Maintain an iterative scope refinement process and preserve flexibility for emerging technical challenges*
Financial	*Component Cost Volatility*	*Discovered significant price fluctuations in specialized hardware components*	*Med*	*Implement 15-20% budget contingency and establish relationships with multiple suppliers*
Team	*Role Definition Clarity*	*Initial responsibility overlap caused minor coordination inefficiencies*	*Low*	*Define clear technical ownership boundaries while maintaining a collaborative integration approach*

Lessons Learned

Technical Lessons

System Integration Complexity: One of the most significant lessons was the complexity of integrating hardware and software systems into a cohesive product. We underestimated the challenges of ensuring seamless communication between the scanner hardware, 3D reconstruction algorithms, cloud infrastructure, and web application. We learned that early integration testing is crucial, waiting until late in the development cycle resulted in discovering compatibility issues that required substantial rework. In future projects, we would implement continuous integration from the earliest stages, even with incomplete subsystems.

Performance Optimization Trade-offs: Balancing reconstruction accuracy with processing speed presented continuous challenges. Our initial algorithm produced accurate models but required over 10 minutes of processing time, exceeding our 300-second target. We learned that optimization requires fundamental algorithm selection and architectural decisions, not just faster code.

Project Management Lessons

Scope Refinement is Continuous: Our initial vision included holographic displays, AR projections, and POS integration. Progress Review 1 feedback made clear we needed to focus on core functionality to deliver a complete, working system. Learning to distinguish between "must-have" and "nice-to-have" features was challenging but essential. The House of Quality analysis proved invaluable for making these decisions objectively based on customer requirements rather than personal preferences.

Test Plans Drive Design Clarity: Developing the comprehensive test plan for Activity 5 forced us to precisely define what "success" meant for each feature. Writing specific, measurable pass/fail criteria revealed ambiguities in our requirements. We learned that defining validation criteria before building ensures you're building the right thing.

Course Reflection

Engineering is About Trade-offs: Perhaps the most valuable lesson from ENSC 440 was understanding that engineering fundamentally involves managing trade-offs. Every decision, camera resolution, cloud provider, material selection, balanced competing objectives: cost vs. performance, simplicity vs. functionality, time-to-market vs. feature completeness. The House of Quality analysis and KPI development provided frameworks for making these trade-offs systematically. We learned there are no "correct" answers in engineering, only well-reasoned choices with understood consequences.

The Value of Constraints: Initially, course constraints—tight deadlines, limited budget, shelf-ready requirement, felt restrictive. However, we came to appreciate that constraints force creativity and prioritization. The budget constraint prevented over-engineering and kept our solution commercially viable. The shelf-ready requirement meant we couldn't hide behind "proof of concept" excuses—every feature had to work reliably. These constraints made us better engineers by forcing us to solve real problems with real limitations.

Interdisciplinary Thinking: RepliVision required knowledge spanning mechanical design, computer vision, cloud computing, web development, user experience, food safety regulations, and business strategy. No single team member possessed all these skills, forcing us to learn from each other and integrate diverse perspectives. This reinforced that complex products require interdisciplinary teams and that effective engineers must communicate across specializations.

From Idea to Product is a Long Journey: Transitioning from ENSC 405W's preliminary design to 440's working product revealed the vast distance between concept and reality. Our 405W design looked impressive but contained dozens of unresolved challenges and overlooked requirements. Actually, building and validating the system exposed every gap in our thinking. We learned that prototyping, testing, and iterating are not optional refinements, they are how you discover what your product needs to be.

Future Works

Immediate Technical Improvements

Enhanced Reconstruction Accuracy: While achieving accuracy across most food types, certain categories present ongoing challenges. Highly reflective surfaces (glazes), transparent elements (beverages), and monochromatic dishes occasionally produce sub-optimal reconstructions. Future work should implement polarized lighting to reduce specular reflections, multi-exposure HDR imaging for extreme contrast situations, and machine learning enhancement trained on thousands of food scans to intelligently fill gaps without hardware changes.

Faster Processing Pipeline: Current processing averages 75-85 seconds, meeting but not exceeding our 90-second target. Optimization opportunities include GPU acceleration (could reduce processing 50-70% to 30-40 second cycles), progressive uploading (beginning cloud upload while reconstruction continues to save 15-20 *seconds), and on-device processing with more powerful embedded processors to eliminate cloud latency entirely while enabling offline operation.*

Feature Additions

Advanced Restaurant Analytics: Add customer engagement metrics tracking which dishes receive most 3D views and longest interaction times, conversion analytics correlating views with actual orders when integrated with POS systems, A/B testing capability for multiple presentations of the same dish, and seasonal trend analysis helping restaurants make data-driven menu decisions.

Enhanced Intelligence: Implement multi-language automatic translation for international markets, ingredient recognition using computer vision to identify common allergens, nutritional estimation combining ingredient recognition with portion size from 3D geometry, and voice-controlled operation for hands-free scanning in busy kitchen environments.

System Integration Opportunities

Point-of-Sale Integration: Enable seamless ordering where customers viewing 3D menus can add items directly to orders syncing with POS systems, dynamic pricing automatically updating catalogue prices when changed in POS, and sales correlation linking 3D menu views with order data for actionable business intelligence.

Kitchen Display System Integration: Display reference 3D models on kitchen screens so chefs can match exact presentations customers viewed when ordering and implement quality control comparing finished plates to 3D models before serving.

Business Development Priorities

Pilot Program Deployment: Deploy 5-10 units with diverse restaurant types (fine dining, fast-casual, ethnic cuisine) to gather real-world usage data, develop case studies documenting measurable impacts on customer satisfaction and operational efficiency, and use pilot feedback for iterative refinement before mass production.

Intellectual Property Protection: File utility patents covering multi-camera dome scanning approach for food and specific algorithms for reconstruction and texture mapping. Register "RepliVision" trademark for international markets. Maintain proprietary algorithms as trade secrets with code obfuscation and security measures.

Agreement

We the undersigned hereby agree to the terms and conditions set out in this charter. No changes can be made to the charter without the agreement of all parties.

Project Sponsor	*Wayne*	
Project Manager	*Mehrshad Azizinia*	Azizinia

House of Quality

To see the completed House of Quality matrix, refer to Appendix A on Page 16.

Our process for developing the HOQ began with identifying the customer requirements (CRs), since all engineering decisions ultimately support them. We started by brainstorming a comprehensive list of potential requirements without weighting them, to capture every possible need. Then, we prioritized and assigned importance ratings based on the system's purpose and restaurant-specific considerations.

The highest-weighted requirement was reconstruction accuracy, as this directly reflects the primary function of the RepliVision system, producing realistic and reliable 3D food representations. The second-highest was compactness and ease of use, since a device that is inconvenient or bulky to operate could discourage adoption despite its benefits. We also included a restaurant-specific requirement for food safety, recognizing that scanning is likely to occur near food preparation areas where hygiene is essential.

Once the CRs were finalized, we derived the Engineering Specifications (ESs) by considering each subsystem of the product (mechanical, electrical, and computational). For each, we identified quantifiable metrics that correspond to customer expectations. For instance, to address the CR of compactness, the associated ES is the dome diameter — a directly measurable physical constraint that influences usability and space efficiency.

During this process, we identified several trade-offs among requirements. Many features have a natural conflict with affordability. For example, ensuring the device is easy to clean and aesthetically pleasing requires additional housing for internal electronics, increasing both cost and heat insulation, which can affect durability. Similarly, pursuing compactness restricts the choice and placement of cameras, capture region and ultimately influencing reconstruction accuracy. These trade-offs were carefully considered and reflected in the relationship and correlation matrices of the HOQ.

In summary, the HOQ revealed that the most critical engineering focus areas are reconstruction accuracy, hygiene, and usability, as they have the strongest relationships with customer satisfaction. Other parameters, such as durability and production cost, play supportive roles but are secondary to the device's ability to consistently generate accurate 3D models. This prioritization will guide subsequent design, testing, and optimization phases for the RepliVision product.

Test Plan

To see the completed Test Plans, refer to Appendix B on Pages 17-20.

Introduction

This test plan establishes systematic validation procedures for the RepliVision 3D food scanning and web catalogue. Testing will verify that all functional requirements are met according to project scope, validate performance against established KPIs and ensuring usability meets restaurant operational standards

Test Plan ID: *TP-REPLIVISION-2025-001*

Testing Scope

The testing scope encompasses all important components of the RepliVision system. This includes the scanner hardware with its cameras, lighting arrays, and structural assembly, the 3D reconstruction and processing pipeline that transforms captured images into digital models, cloud storage and data management infrastructure for reliable model hosting; the web-based catalogue and viewing interface that presents dishes to customers; and the operator interface along with supporting documentation that enables restaurant staff to use the system effectively

Testing Types

Functional Testing: Verify features operate according to specifications

Performance Testing: Measure quantitative metrics against KPI targets

Usability Testing: Evaluate user experience with representative users

Compatibility Testing: Validate functionality across platforms and devices

Pass/Fail Criteria:

The system will be deemed ready for deployment if all mandatory functional tests pass without critical failures, 90% or more KPIs meet or exceed their target values, no critical defects such as system crashes, data loss, or safety hazards are identified, usability ratings average at least 4.0 out of 5.0 from test participants, and documentation enables 90% or higher task completion success rates without external assistance.

Progress Review 1

RepliVision Inc.
Company 21
MINUTES 1
September 19, 2025
2:30 – 3:20 P.M.
SFU ASB L-9014

Present: *Mehrshad Azizinia, Curtis Huang, Shervin Khamooshian, Joseph Wen*
Absent: *None*
Purpose of Meeting: *To show our progress after the transition from 405W, present and explain our project, and review the deliverables and project objectives as we move forward in 440.*
Minutes:
The meeting was called to order late at 2:30 P.M. due to project setup. Curtis agreed to record the minutes.

A. ***Approval of the minutes of the September 19, 2025 meeting***

B. ***Project Overview ~ 6min***

   Discussion: *Mehrshad provided an overview of the problem statement, solution, and demonstrated the project's features and functions.*
   Action*: None*

C. ***Scope Overview ~ 38min***

   Target Customer

   Discussion: *Professor Jannesar asked who the project was being marketed to. Mehrshad confirmed that fine dining restaurant owners are the intended customers. It was noted that the concept may reflect customer desires more than restaurant owner needs.*

   Action:
   - *Interview restaurant managers and owners to gather opinions.*
   - *Showcase both the Replicator and HoloVision during interviews.*
   - *Curtis to contact two local businesses; Mehrshad to contact one local business.*

   System Practicality

Discussion: *Concerns were raised that the display system may be too large or impractical for restaurant use. Questions were also raised about whether the scanner system might become a one-time-use item. No resolution was reached, and the issue remains open for further input.*

Action:

- *Incorporate questions on size, practicality, and repeated use into upcoming interviews with restaurant managers and owners.*

D. ***Next Meeting Date***
The next meeting will be scheduled between October 27-31, 2025.

E. ***Other Business***
Action*: Joseph and Shervin will research techniques and requirements for converting scans into mesh instead.*

Adjournment
The meeting was adjourned at 3:20 P.M. (10 minutes late due to extended discussion).

Progress Review 2

RepliVision Inc.

Company 21

MINUTES 2

October 25, 2025

2:30 – 3:00 P.M.

SFU ASB 9705

Present: *Mehrshad Azizinia, Curtis Huang, Shervin Khamooshian, Joseph Wen*

Absent: *None*

Purpose of Meeting: *To show our progress after Progress Review Meeting 1, present and explain our updated project, and review the deliverables and project objectives as we approach the end of 440.*

Minutes:

The meeting was called to order at 2:30 P.M. Curtis agreed to record the minutes.

A. ***Approval of the minutes of the October 25, 2025 meeting***

B. ***Project Overview ~ 32 min***

Scanner System ~ 27 min

Discussion: *Mehrshad provided an overview of previous design scanner system and shown new design of scanner. Mehrshad mentioned certain challenges we are facing regarding the scanner system.*

Professor Jannesar said, "You are short on time, so just begin making the new scanner." Professor Jannesar also gave some advice for the challenges we were facing.

***Action**: The team will proceed with finish designing and start constructing the new scanner based on the updated design. Mehrshad will lead the implementation and address the identified challenges using the professor's suggestions. Progress will be reviewed in the next meeting.*

System Pipeline ~5 min

***Discussion:** Mehrshad discussed the system pipeline and explained how cloud storage was integrated into the workflow. The team reviewed the current data flow between the scanner, the reconstruction module, and the web interface.*

***Action:** Curtis, Joseph, and Shervin will update the web application, perform code refactoring to improve maintainability, and conduct comprehensive system testing to ensure full integration and reliability.*

C. ***Next Meeting Date***
The next meeting will be scheduled on October 30, 2025, for progress update.

D. ***Other Business***
***Action**: None*

Adjournment

The meeting was adjourned early at 3:02 P.M.

Progress Review 3 (Final Demo)

RepliVision Inc.

Company 21

MINUTES 3

November 27, 2025

4:00 – 4:40 P.M.

SFU ASB Hallway

***Present:** Mehrshad Azizinia, Curtis Huang, Shervin Khamooshian, Joseph Wen*

***Absent:** None*

***Purpose of Meeting:** To showcase the completed RepliVision system to the instructional team and external testers, validate shelf-readiness through comprehensive testing, and demonstrate compliance with project scope and KPIs.*

Minutes:

The meeting was called to order at 4:00 P.M.

A. Approval of the minutes of the November 27, 2025 meeting

Meeting was approved as amended:

- *Meeting location changed from ASB 9705 to ASB Hallway*

B. Project Sales Pitch ~ 1 min

***Discussion:** Mehrshad delivered the one-minute product pitch to the instructional team and testers. The instructional team and testers expressed strong interest in the product and its practical applications for the restaurant industry and offers $900 for the product.*

***Action:** None*

C. Operating Manual and Testing ~ 38 min

***Discussion:** The product was delivered to the instructional team and testers with all required components and documentation. Testers successfully unpacked, installed, and tested the system following only the operation manual without team intervention.*

***Action:** None*

D. Demo Conclusion ~4 min

***Discussion:** Professor Jannesar provided final feedback, commending the team for completing the project, when the team was in critical condition during Progress Review 2. The demo ended with the instructional team and the testers deeming this functional and is shelf ready. He mentioned adding 2 additional sections to the final report, including lessons learned and future works.*

***Action:** To complete the final report by integrating the deliverable in the requirements along with the additional sections mentioned by the professor.*

E. Other Business
***Action**: None*

Adjournment

The meeting was adjourned early at 4:40 P.M.

Appendix A: House of Quality Chart

Project:	RepliVision
Date:	12-Oct-25

Engineering Specification

Dome Material Strength (MPa)																		
Dome reflectiveness (%)	+																	
Dome diameter (cm)																		
Device total weight (kg)	+		+															
Structural vibration damping factor (ratio)	+		-	-														
Durability (cycles)	+			-	+													
Power Consumption (W)				+														
Camera resolution (MP)		+																
Illumination uniformity (%)		+					-	+										
Scan cycle time (s)				-			-											
Raspberry Pi operating temperature (°C)							+											
LED lifetime (hours)							-											
3D reconstruction accuracy (mm error)								+	+	+								
File processing time (s)										-			+					
Data storage per scan (MB)													+					
Web interface load time (s)																		
Menu system load time (%)																		
Auto calibration success rate (%)													+					
Direction of Improvement	□	▲	□	▼	▲	▲	▼	▲	▲	▼	▼	▲	▼	▼	▼	▼	▼	▲

Customer Competitive Assessment

Relative Weight	Restaurant Owners	Customer Requirements	Dome Material Strength (MPa)	Dome reflectiveness (%)	Dome diameter (cm)	Device total weight (kg)	Structural vibration damping factor (ratio)	Durability (cycles)	Power Consumption (W)	Camera resolution (MP)	Illumination uniformity (%)	Scan cycle time (s)	Raspberry Pi operating temperature (°C)	LED lifetime (hours)	3D reconstruction accuracy (mm error)	File processing time (s)	Data storage per scan (MB)	Web interface load time (s)	Menu system load time (%)	Auto calibration success rate (%)	RepliVision	Phone-based 3D Models	Wax Models	Standard Photo Menus
12.90%	4	Easy to operate	▽	▽	o	●						▽					▽	▽	o	o	5	4	2	5
16.13%	5	Accurate 3D representation		●	o		●	●		●	●	o			●		o			▽	4	3	4	1
9.68%	3	Fast scanning and processing		▽								●	▽		o	●	●	▽			3	3	2	5
12.90%	4	Compact and fits restaurant counters			●	o															4	5	1	5
9.68%	3	Durable and reliable hardware	●				●	●	▽					●							4	5	2	5
9.68%	3	Easy to clean and maintain	o		o	o		o													4	3	5	5
9.68%	3	Safe and hygienic for food use	o		▽								▽								5	5	3	5
6.45%	2	Affordable price	o	▽		o			●	●	o			▽			▽		□		4	5	2	5
6.45%	2	Low power consumption							●			o	▽	▽	▽						5	4	1	5
6.45%	2	Aesthetically pleasing design	▽	o																	5	2	5	4
		Importance Rating Sum (Importance x Relationship)	184	194	242	203	232	261.29	126	203	165	168	25.8	100	181	87.1	155	22.6	58.1	54.8				
		Relative Weight	15%	7%	9%	8%	9%	10%	5%	8%	6%	6%	1%	4%	7%	3%	6%	1%	2%	2%				
		Target Value	≥ 40	100	≤ 280	≤ 3	≤ 30	≥ 10,000	≤25	≥12	≥95	≤30	≤60	≥ 10,000	≤1.0	≤20	≤50	≤2	≥99	≥95				
		RepliVision	5	5	4	4	5	5	4	5	5	4	5	5	5	4	4	4	5	5				
		Phone-based 3D Models	-	-	-	-	-	-	5	3	3	4	-	-	3	3	3	-	-	3				
		Wax Models	-	-	-	-	-	3	5	3	-	-	-	-	4	-	-	-	-	2				
		Standard Photo Menus	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5				

Technical Competitive Assessment

Correlations	
Positive	+
Negative	-
No Correlation	

Relationships		Weight
Strong	●	9
Medium	o	3
Weak	▽	1

Direction of Improvement	
Maximize	▲
Target	□
Minimize	▼

Appendix B: Test Cases

Functional Test

Date:	***Test Case ID:*** *TC-FUNC-001*
Test Objective: *Verify scanner hardware components function correctly and synchronously*	
Methodology: *1. Power on scanner system 2. Verify LED lighting activates uniformly 3. Check status light 4. Place test dish in scanner 5. Trigger scanning sequence 6. Verify automatic image transfer*	
Expected Results: *All LEDs illuminate within ≤ 1 second, All cameras are operational, Capture completes without errors, Images transfer with zero corruption, Interface displays "Ready" light once again.*	
Pass/Fail Criteria: Pass*: All expected results achieved* ***Fail****: Any camera fails or image corruption*	
Actual Results:	

Date:	***Test Case ID:*** *TC-FUNC-002*
Test Objective: *Validate geometric accuracy and visual fidelity of 3D reconstructions*	
Methodology: *Perceptually gauge if it looks representative enough (exceeds user threshold of if "enough") from all viewing angles*	
Expected Results: *Most scans look correct*	
Pass/Fail Criteria: Pass: *User believes the model is sufficiently representative* ***Fail:*** *Model does not accurately depict the actual object*	
Actual Results:	

Date:	***Test Case ID:*** *TC-FUNC-003*
Test Objective: *Confirm reliable model upload, storage, and retrieval from cloud*	
Methodology: *1. Complete scan of test dish 2. Monitor upload process 3. Record upload timestamps 4. Verify progress indicator accuracy 5. Confirm success message 6. Access catalogue on separate device 7. Download model and compare hash 8. Simulate network interruption 9. Test retry mechanism 10. Upload 10 consecutive models*	
Expected Results: *Upload completes within 120 seconds. Model in catalogue within 180 seconds. Also network error triggers clear message and retry mechanism works*	
Pass/Fail Criteria: Pass*: ≥90% upload success, error handling functional, no corruption* ***Fail****: Success rate <90% or data integrity issues*	
Actual Results:	

Date:	***Test Case ID:*** *TC-FUNC-004*
Test Objective: *Validate interactive web application functionality*	
Methodology: *1. Access web application on 2. Measure homepage load time 3. Navigate menu categories 4. Select dish and measure model load time 5. Test 3D viewer controls (rotate, zoom, reset) 6. Test management features (add/edit/remove)*	
Expected Results: *Homepage loads within 5 seconds, category navigation < 1s, 3D model displays within 30 seconds, touch controls respond immediately (60 FPS), all management functions work, changes persist after refresh*	
Pass/Fail Criteria: Pass*: Load times meet targets, all controls functional, 100% management features work* ***Fail:*** *Load time >20s, controls unresponsive, or management function fails*	
Actual Results:	

Performance Test

Date:	***Test Case ID:*** *TC-PERF-001*
Test Objective: *Measure complete workflow duration from scan initiation to customer-ready model*	
Methodology: *1. Prepare 2 test dishes (varied complexity) 2. For each dish: Record start time when placed in scanner, initiate scan, allow automatic processing, record end time when model visible in catalogue, calculate duration, repeat 3 times per dish. Calculate statistics (average, min, max, std dev)*	
Expected Results: *Average completion time ≤200 seconds, 90% of scans complete within 300 seconds, no single scan exceeds 420 seconds, standard deviation ≤15 seconds, consistent performance across complexity levels*	
Pass/Fail Criteria: Pass*: Average ≤200s, 90%+ complete within 300s* ***Fail****: Average >300s or >10% exceed 300s*	
Actual Results:	

Date:	***Test Case ID:*** *TC-PERF-002*
Test Objective*: Evaluate web application performance under multiple simultaneous users*	
Methodology: *1. Run 5 concurrent users access app, moniter server response times, record average load time 3. Analyze server logs for errors*	
Expected Results: *Application responsive with 5 users, load times stay <6 seconds, zero crashes or error states, no significant degradation from 5 to 10 users*	
Pass/Fail Criteria: Pass*: Successful operation with 5 users, performance targets met* ***Fail****: Crashes, timeouts, or load times >5s*	
Actual Results:	

Usability Test

Date:	*Test Case ID: TC-USAB-001*
Test Objective: *Validate system learnability and ease of use for restaurant staff*	
Methodology: *1. Provide 15-minute training: Scanner safety (3 min), scan cycle demo (7 min), interface review (3 min), Q&A (2 min) 2. For each trainee: Provide task and documentation only, start timer, observe without intervention, record completion time and success*	
Expected Results: *100% complete successful scan, average completion time ≤10 minutes, no safety incidents, documentation referenced effectively*	
Pass/Fail Criteria: Pass*: All trainees succeed, avg time ≤10 min* ***Fail****: Any trainee fails*	
Actual Results:	

Date:	*Test Case ID: TC-USAB-002*
Test Objective: *Assess customer-facing menu interface for intuitiveness and satisfaction*	
Methodology: *1. Provide minimal context (no instruction) 2. For each tester: Observe interaction silently, record time to first model view, note navigation patterns 3. Conduct post-test interview. 4. Compile quantitative and qualitative feedback*	
Expected Results: *90%+ view 3D model without assistance, Time to first view < 30 seconds, average confidence rating ≥ 4.0/5.0, average visual quality rating ≥4.0/5.0, majority prefer over photo menus, 85%+ agree models accurate*	
Pass/Fail Criteria: Pass*: 90%+ success, confidence ≥4.0, quality ≥4.0* ***Fail****: Success rate <90% or ratings <4.0*	
Actual Results:	

Date:	*Test Case ID: TC-USAB-003*
Test Objective: *Verify cleaning procedures meet restaurant hygiene standards*	
Methodology: *1. Contaminate dome with simulated residue 2. For 3 test subjects: Provide sanitation protocol, time cleaning procedure, visual inspection under bright light, Run scanner function test, inspect for moisture ingress 3. Calculate average cleaning time 4. Document issues/suggestions*	
Expected Results: Complete sanitation within 5 minutes, no visible residue, scanner functions normally, no moisture damage, consistent results across subjects	
Pass/Fail Criteria: Pass*: Avg time ≤5 min, scanner operational* ***Fail****: Time >5 min or malfunction*	
Actual Results:	

Compatibility Test

Date:	*Test Case ID: TC-COMP-001*
Test Objective: *Assess scanner performance across diverse food categories*	
Methodology: *1. Test with varied food types: Dry/matte (bread, vegetables), glossy/reflective (sauces, glazes), transparent (beverages, jellies), complex geometry (garnishes, layers), dark/light extremes (chocolate, cream) 2. For each: scan using standard procedure, visual quality assessment, document any challenges3. Test 8" and 10" plate sizes*	
Expected Results: *Acceptable quality (≤5mm eyeballed) for 80% of types, both plate sizes captured completely, Plate diameter within ±2mm, challenging categories documented*	
Pass/Fail Criteria: Pass*: Both sizes meet accuracy, 80%+ food types acceptable* ***Fail****: Either size fails or <80% food types acceptable*	
Actual Results:	